

1. Ans: a)

$$200 + x + 300 = 20,000,500$$

$$\Rightarrow x = 20,000,500 - (300 + 200)$$

$$= 20,000,000 \text{ ns}$$

$$= 20,000,000 \times 10^{-9} \text{ s}$$

$$20,000,000 \times 10^{-9} \text{ s} \quad \frac{2048 \text{ byte}}{20,000,000 \times 10^{-9} \text{ s}}$$

$\therefore 1 \text{ s}$

$$\frac{2048}{20,000,000 \times 10^{-9}}$$

1 Byte/s

$$= 102400 \text{ Byte/s}$$

$$= \frac{102400}{2^{10}} \text{ Kb/s}$$

$$= 100 \text{ Kb/s (Ans)}$$

$$\textcircled{b} \quad 2048 \div 2 = 1024$$

Brust mode

$$200 + \frac{1024 \times 10^9}{100 \times 2^{10}} + 300 = 10,000,500 \text{ ns}$$

cycle stealing mode

$$\left(200 + \frac{4 \times 10^9}{100 \times 2^{10}} + 300 \right) \times \frac{1024}{4} = 10,128,000 \text{ ns}$$

Total time needed:

Brust mode + cycle stealing mode

$$= (10,000,500 + 10,128,000) \text{ ns}$$

$$= 20,128,500 \text{ ns (Ans)}$$

2. Ans: a)

$$\begin{array}{r} 7A34 \text{ H} \\ + 4DC2 \text{ H} \\ \hline C7F9 \text{ H} \end{array}$$

$$\begin{array}{r} + 0111 \quad 1010 \quad 0011 \quad 0100 \\ + 0100 \quad 1101 \quad 1100 \quad 0010 \\ \hline (-) 1100 \quad 0111 \quad 1111 \quad 1001 \end{array}$$

SF = 1 (As result is negative, msb is 1)

PF = 1 (Even number of '1's in low 8 bit of result)

CF = 0 (As there is no carry)

OF = 1 ((+) + (+) must be (+) result, but our result is (-))

2(b) A.B = 24 bit, D.B = 8 bit.

$$\text{Total memory} = (2^{24} \times 8) \text{ bit}$$

$$= 2^{24} \text{ byte}$$

$$= 2^{14} \text{ KB}$$

$$\therefore \text{Number of segments} = \frac{2^{14} \text{ KB}}{128 \text{ KB}}$$

$= 128 \text{ segments}$
(Ans)

3. (a) $DS = 4C18H$ $BX = 0100H$

Physical address = $(4C18 \times 10)H + 0100H$

$= 4C280H$

I) $PA - offset$

$= 4C280H - B290H$

$= 40FF0$ (valid) as there is 0 at last

$\therefore \text{Segment register} = \frac{40FF0H}{10H}$

$= 40FFH$ (valid)

(ii) $Offset = PA - (\text{segment} \times 10H)$

$= 4C280 - (3C2F \times 10)$

$= FF90H$ (valid)

$$\text{iii) First PA} = \text{E47B0 H}$$

$$\therefore \text{Last PA} = \text{E47B0 H} + \text{FFFF}$$

$$= \text{F47AF H (Ans)}$$

2/
1. Ans (a)

RAM[A]

Data bus = 32 bit

RAM[B]

A.B = 20 bit

D.B = 8 bit

RAM[B]

$$\therefore \text{Ram size} = 2^{20} \times 8 = \frac{2^{20} \times 8}{2^{10} \times 8} \text{ KB}$$
$$= 1024 \text{ KB}$$

RAM[A]

Ram size = 1024 KB

$$\therefore 1024 \times 2^{10} \times 8 = 2^x \times 32$$

$$\therefore x = \log_2 \left[\frac{1024 \times 2^{10} \times 8}{32} \right]$$

= 18 bit

(P.T.O)

$\therefore$ Address bus of RAM[A] = 18 bit (Ans)

Total memory capacity of both
RAMS = 1024 KB (Ans)

2.

a)

MP

A.B = 18 bit

D.B = 8 bit

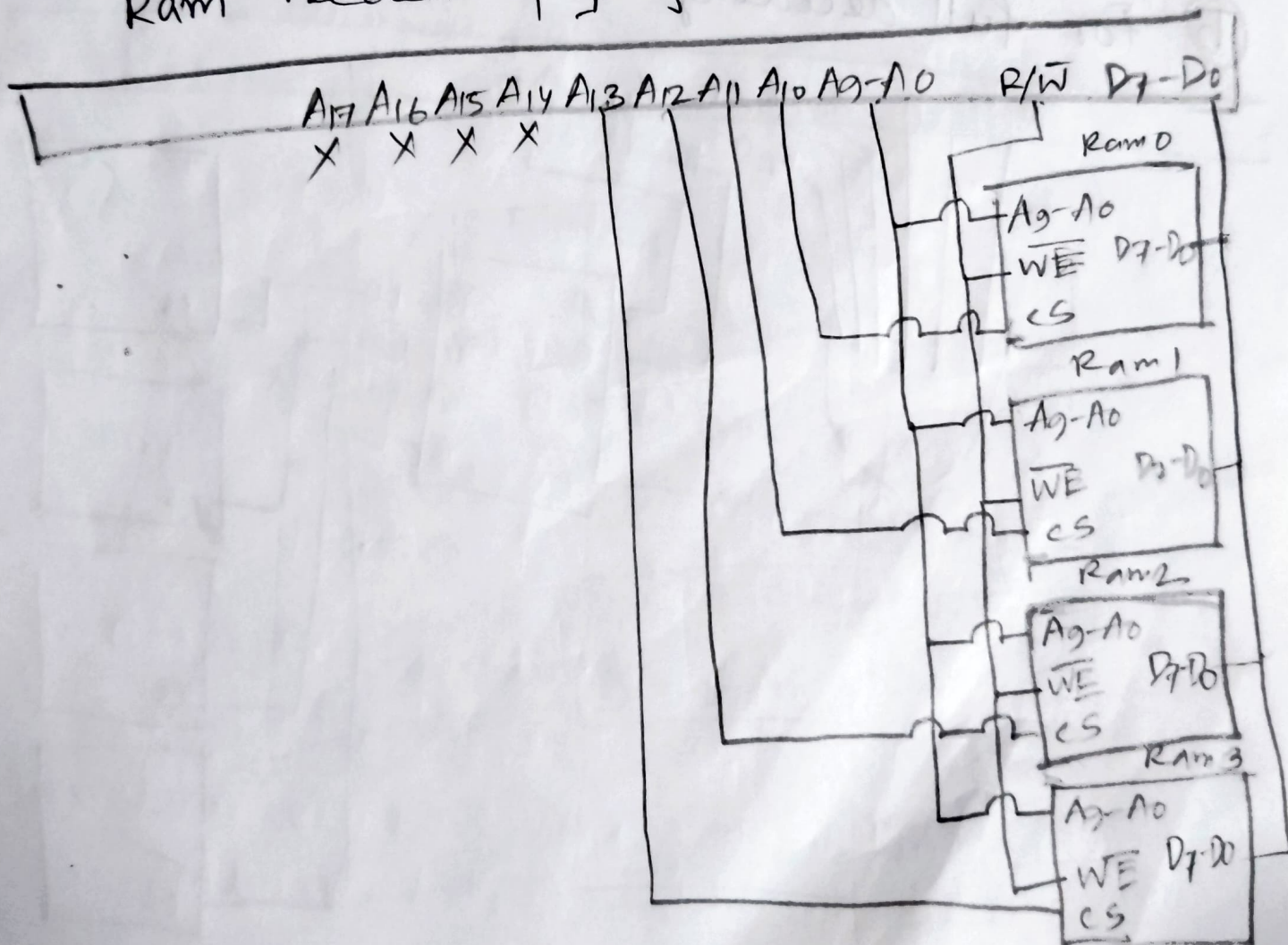
RAM

A.B = 10 bit

D.B = 8 bit

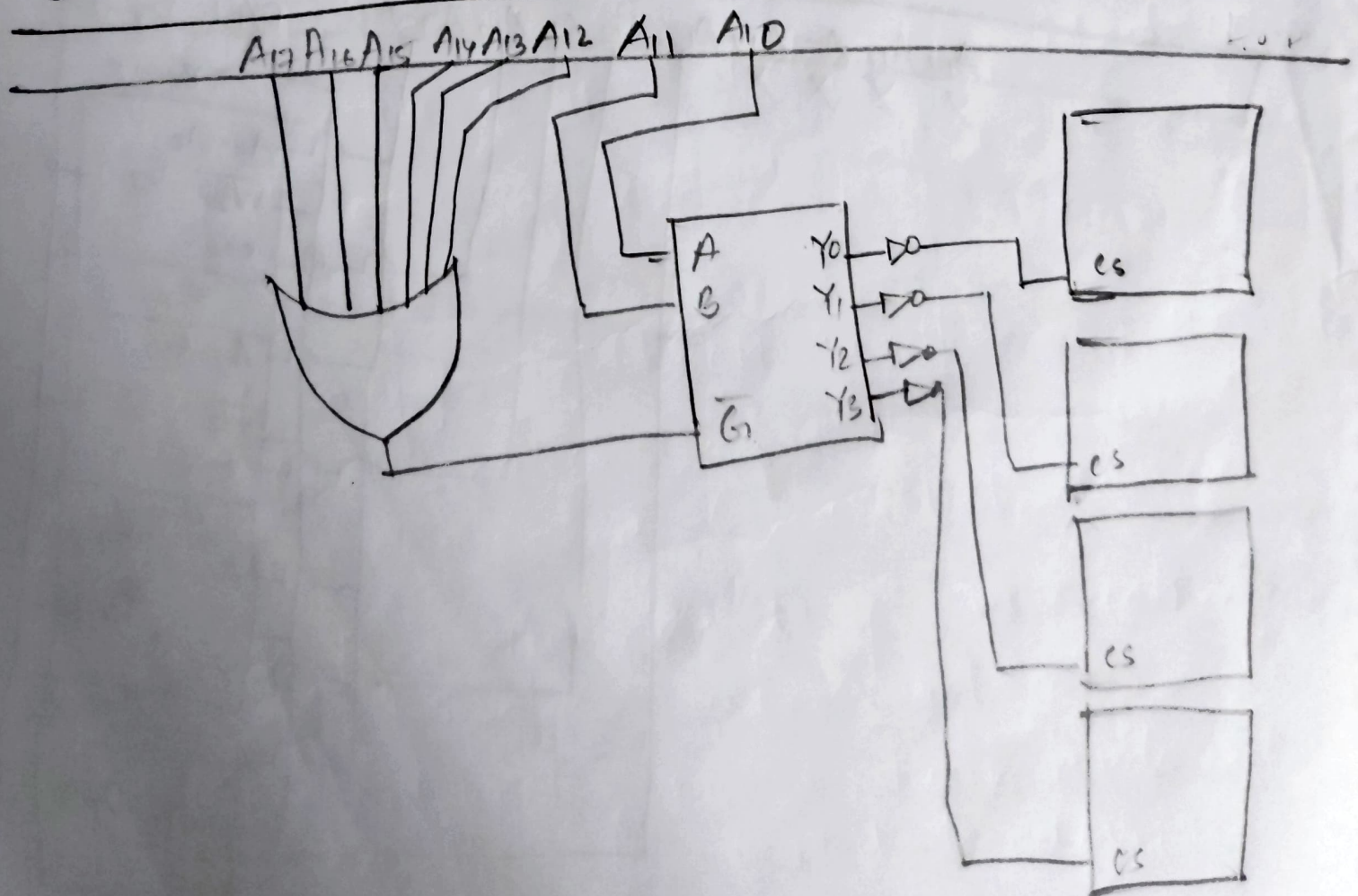
$$\text{RAM capacity} = \frac{2^{10} \times 8}{2^{10} \times 8} = 1 \text{ KB}$$

$$\text{Ram needed} = \left\lceil \frac{4}{1} \right\rceil = 4 \text{ Ram}$$



4				10	Address map	
$A_{17} - A_{14}$				$A_9 - A_0$	Ram chip	
0000	0	0	0	1	0 - - - 0	00400 H
0000	0	0	0	1	1 - - - 1	007FF H <i>Ram0</i>
"	0	0	1	0	0 - - - 0	00800 H
"	0	0	1	0	1 - - - 1	00BFF H <i>Ram1</i>
"	0	1	0	0	0 - - - 0	01000 H
"	0	1	0	0	1 - - - 1	013FF H <i>Ram2</i>
"	1	0	0	0	0 - - - 0	02000 H
"	1	0	0	0	1 - - - 1	023FF H <i>Ram3</i>

⑥ For full decoding 4:2 decoder is needed



Address map		A ₁₁ A ₁₀		A ₉ - A ₀		A ₁₇ - A ₁₂	
00000H	000000	0	0	0	0	0	0
003FFH	000000	0	0	1	1	0	0
00400H	000000	0	0	0	0	0	0
007FFH	000000	0	0	1	1	0	0
00800H	000000	0	0	0	0	0	0
00BFFH	000000	0	0	1	1	0	0
00C00H	000000	0	0	0	0	0	0
00FFFH	000000	0	0	1	1	0	0

In Data Collision for hardware fault in may occur in partial decoding but in full decoding no collision is not occur.

In memory (address) not unique for 1 location and it wastes memory space for partial decoding but in full decoding no address location has unique address but for full decoding more hardware needs but compare to partial decoding.

Compares Full decoding is better for the reason below;

Comparison of partial and full decoding methods

// Waste of address space in partial decoding occur, but in full decoding no waste of address space.

// Address map is not contiguous in partial decoding, it is sparsely distributed, but in full decoding address map is contiguous.

// Data collision for hardware fault may occur in partial decoding, but in full decoding data collision is not occur.

// Memory foldback (address not unique for 1 locations) and it wastes memory space for partial decoding but in full decoding each memory location has unique address

// For full decoding more hardware needs but compare to partial decoding.

3. (a)

$CS = 0080H$, $IP = 00FFH$, flag register $= FFFFH$

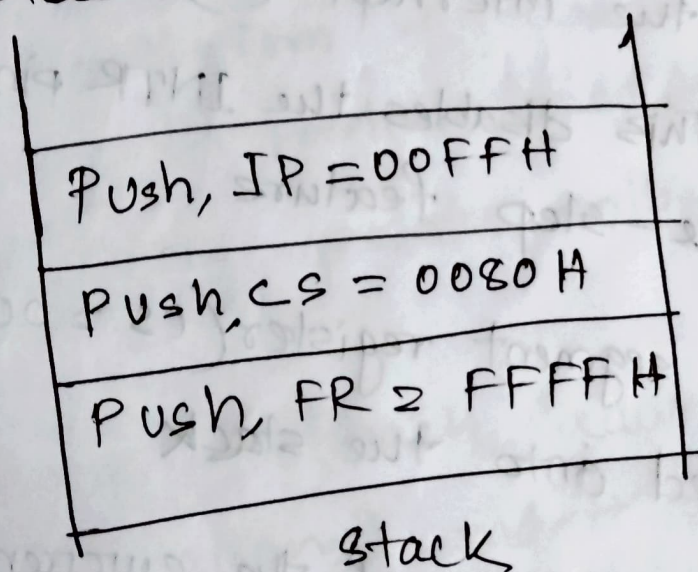
Sequence of events when encounters a software interrupt:

- i) Push flag - $FFFFH$ onto the stack
- (ii) Both the interrupt (IF) & trap (TF) are set to 0. This disables the INTR pin & the trap or single-step feature
- (iii) Code segment register ($CS = 0080H$) are pushed onto the stack
- (iv) The contents of the current instruction pointer ($IP = 00FFH$) are pushed onto the stack

(v) The interrupt vector contents are fetched,
& then placed into both (IP = 00FFH) &

(CS = 0080H) so that, the next
instruction executes at the interrupt
service procedure addressed by the
interrupt vector

Stack PUSH



Stack POP

Pop, IP = 00FFH
Pop, CS = 0080H
Pop, FR = FFFFH

stack

3.(b) Overflow interrupt (10H - 13H)
Floating point Error (40H - 43H)

Interrupt number of overflow interrupt = 4

Interrupt " " floating point error = 10

∴ Overflow interrupt has higher priority,
because, the lower the vector number,
the higher the priority